



BKP: An R Package for Beta Kernel Process Modeling

Jiangyan Zhao

School of Statistics, ECNU

jyzhao@sfs.ecnu.edu.cn

joint work with Kunhai Qing and Jin Xu

2026.08.08

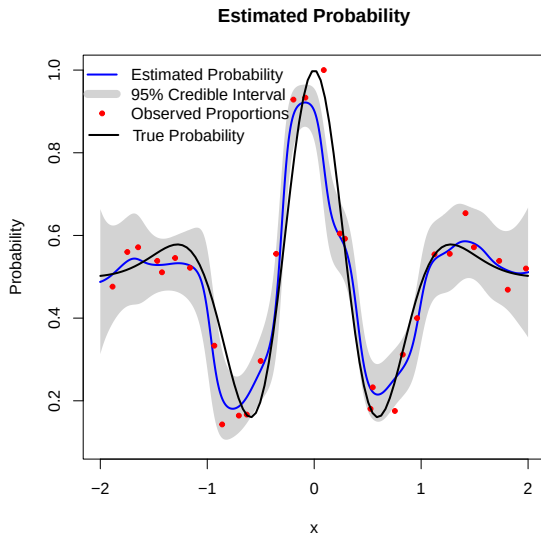


Contents

- 1 Motivation
- 2 Methodology
- 3 Package Usage
- 4 Examples and Applications
- 5 Extensions
- 6 Summary



Estimating an Unknown Probability Surface



- **Observation model**

$$y_i \mid \pi(\mathbf{x}_i) \sim \text{Binomial}(m_i, \pi(\mathbf{x}_i)).$$

- **Objective**

Estimate the underlying probability surface $\pi(\mathbf{x})$ and quantify its pointwise uncertainty.



The Gap Between Flexibility and Tractable Inference

Logistic regression

$$\pi(\mathbf{x}) = \text{logit}^{-1}(\mathbf{x}^\top \boldsymbol{\beta})$$

Fast and interpretable
Limited nonlinear flexibility

Gaussian process classification

$$\pi(\mathbf{x}) = \text{logit}^{-1}\{f(\mathbf{x})\}, \quad f(\mathbf{x}) \sim \text{GP}$$

Flexible probability surfaces
Approximate posterior inference
(Rasmussen and Williams, 2006; Nickisch and Rasmussen, 2008)

Motivation

Can we retain nonlinear flexibility while performing direct, closed-form updating on the probability scale?



Contents

- 1 Motivation
- 2 Methodology
- 3 Package Usage
- 4 Examples and Applications
- 5 Extensions
- 6 Summary



Beta Kernel Process: Direct Probability-Scale Updating

Beta prior

$$\pi(\mathbf{x}) \sim \text{Beta}(\alpha_0(\mathbf{x}), \beta_0(\mathbf{x})).$$

Prior information on the probability scale

+

Local likelihood

$$k_i(\mathbf{x}) = k(\mathbf{x}, \mathbf{x}_i; \theta),$$

$$S(\mathbf{x}) = \sum_{i=1}^n k_i(\mathbf{x}) y_i,$$

$$F(\mathbf{x}) = \sum_{i=1}^n k_i(\mathbf{x}) (m_i - y_i).$$

Nearby observations receive greater weight

$\Rightarrow$

Beta posterior

$$\pi(\mathbf{x}) | \mathcal{D} \sim \text{Beta}(\alpha_n(\mathbf{x}), \beta_n(\mathbf{x})),$$

$$\alpha_n(\mathbf{x}) = \alpha_0(\mathbf{x}) + S(\mathbf{x}),$$

$$\beta_n(\mathbf{x}) = \beta_0(\mathbf{x}) + F(\mathbf{x}).$$

Closed-form posterior inference

Key idea

Local kernel weighting + Beta-binomial conjugacy
 $\Rightarrow$ **flexible modeling with closed-form posterior inference.**

BKP is a **Bayesian-inspired** probability-scale smoother based on kernel-weighted local-likelihood conjugate updating (Goetschalckx et al., 2011; MacKenzie et al., 2014; Rolland et al., 2019; Fan et al., 1998).



Closed-Form Posterior Summaries

Probability estimation

$$\hat{\pi}(\mathbf{x}) = \frac{\alpha_n(\mathbf{x})}{\alpha_n(\mathbf{x}) + \beta_n(\mathbf{x})}$$

Posterior mean of the unknown probability surface

Pointwise uncertainty

$$\text{Var}\{\pi(\mathbf{x}) \mid \mathcal{D}\} = \frac{\hat{\pi}(\mathbf{x})\{1 - \hat{\pi}(\mathbf{x})\}}{\alpha_n(\mathbf{x}) + \beta_n(\mathbf{x}) + 1}$$

$$\text{CI}_{1-\delta}(\mathbf{x}) = [Q_{\delta/2}(\mathbf{x}), Q_{1-\delta/2}(\mathbf{x})]$$

Credible intervals follow from Beta posterior quantiles

Count prediction

$$Y_{\text{new}} \mid \mathcal{D}, \mathbf{x}, m_{\text{new}} \sim \text{Beta-Binomial}\{m_{\text{new}}, \alpha_n(\mathbf{x}), \beta_n(\mathbf{x})\}.$$

Closed-form prediction for future binomial counts

Computational implication

Conditional on the kernel hyperparameters, posterior summaries and predictive distributions are available without numerical posterior approximation.



Effective-Sample-Size Calibration

Idea

- When trial sizes are **heterogeneous**, the standard kernel-weighted data contribution to **posterior precision**,

$$m_K(\mathbf{x}) = \mathbf{k}(\mathbf{x})^\top \mathbf{m},$$

may **misrepresent** the intended amount of local information.

- ESS calibration **rescales** the data pseudo-count to a Shepard-based target (Shepard, 1968) while preserving the kernel-weighted **empirical success proportion**.

Target ESS

$$m_{\text{tar}}(\mathbf{x}) = \rho(\mathbf{x}) m_S(\mathbf{x}),$$

$$\rho(\mathbf{x}) = \max_{1 \leq i \leq n} k(\mathbf{x}, \mathbf{x}_i).$$

$m_S(\mathbf{x})$ **interpolates** the observed trial sizes.

$\rho(\mathbf{x})$ **attenuates** the target ESS in regions with weak data support.

Calibrated update

For $m_K(\mathbf{x}) > 0$, define $c(\mathbf{x}) = \frac{m_{\text{tar}}(\mathbf{x})}{m_K(\mathbf{x})}$.

$$\alpha_n^{\text{ESS}}(\mathbf{x}) = \alpha_0(\mathbf{x}) + c(\mathbf{x}) \mathbf{k}(\mathbf{x})^\top \mathbf{y},$$

$$\beta_n^{\text{ESS}}(\mathbf{x}) = \beta_0(\mathbf{x}) + c(\mathbf{x}) \mathbf{k}(\mathbf{x})^\top (\mathbf{m} - \mathbf{y}).$$



Prior Specification

Non-informative prior

$$\pi(\mathbf{x}) \sim \text{Beta}(1, 1).$$

Default uniform prior when no external information is available

Informative prior with fixed mean

$$\alpha_0(\mathbf{x}) = r_0 p_0, \quad \beta_0(\mathbf{x}) = r_0(1 - p_0)$$

$$\mathbb{E}\{\pi(\mathbf{x})\} = p_0, \quad \alpha_0(\mathbf{x}) + \beta_0(\mathbf{x}) = r_0$$

A fixed **prior mean** p_0 with **precision** controlled by r_0

Data-adaptive prior

$$p(\mathbf{x}) = \sum_{i=1}^n \frac{k(\mathbf{x}, \mathbf{x}_i)}{\sum_{i=1}^n k(\mathbf{x}, \mathbf{x}_i)} \frac{y_i}{m_i}, \quad r(\mathbf{x}) = r_0 \sum_{i=1}^n k(\mathbf{x}, \mathbf{x}_i)$$

$$\alpha_0(\mathbf{x}) = r(\mathbf{x})p(\mathbf{x}), \quad \beta_0(\mathbf{x}) = r(\mathbf{x})\{1 - p(\mathbf{x})\}.$$

The **prior mean** follows the local empirical proportion, while the **prior precision** adapts to the local kernel mass



Kernel Functions and Length Scales

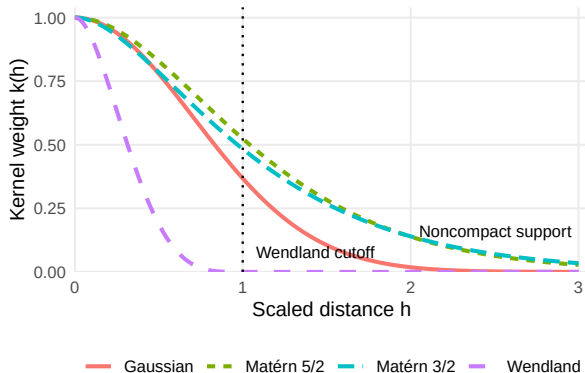
Scaled distance

$$h(x, x'; \theta) = \sqrt{\sum_{j=1}^d \left(\frac{x_j - x'_j}{\theta_j} \right)^2}.$$

Length-scale structure

Isotropic: $\theta_1 = \dots = \theta_d = \theta$

Anisotropic: $\theta = (\theta_1, \dots, \theta_d)$



Borrowing range

The kernel family controls the shape of information decay, whereas θ controls the borrowing range.



LOOCV-Based Length-Scale Selection

Leave-one-out quantities

- $\hat{\pi}^{-i}(\mathbf{x}_i; \boldsymbol{\theta})$: predicted probability at $\mathbf{x}_i$ after excluding observation i .
- $\tilde{\pi}_i = y_i/m_i$: observed empirical proportion.

Selection criterion

$$\hat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta}}{\operatorname{argmin}} L(\boldsymbol{\theta}; \mathcal{D}_n), \quad L \in \{\text{BS}, \text{LL}\}.$$

Brier score

$$\text{BS}(\boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^n \{ \hat{\pi}^{-i}(\mathbf{x}_i; \boldsymbol{\theta}) - \tilde{\pi}_i \}^2.$$

Log-loss

$$\text{LL}(\boldsymbol{\theta}) = -\frac{1}{n} \sum_{i=1}^n \left[\tilde{\pi}_i \log \hat{\pi}^{-i}(\mathbf{x}_i; \boldsymbol{\theta}) + (1 - \tilde{\pi}_i) \log \{1 - \hat{\pi}^{-i}(\mathbf{x}_i; \boldsymbol{\theta})\} \right].$$

All leave-one-out predictions are obtained from one kernel matrix by setting its diagonal entries to zero.



Multi-Start Hyperparameter Optimization

Log-scale reparameterization

$$\gamma_j = \log_{10}(\theta_j), \quad \theta_j = 10^{\gamma_j}.$$

$$d_\gamma = \begin{cases} 1, & \text{isotropic,} \\ d, & \text{anisotropic.} \end{cases}$$

The transformation enforces positive length-scale parameters

Optimization workflow

- 1 Generate $n_0 = 10d_\gamma$ initial values using a Latin hypercube design
- 2 Run the derivative-free SBPLX algorithm from each initial value
- 3 Retain the solution with the smallest LOOCV loss

$$s^* = \operatorname{argmin}_{1 \leq s \leq n_0} L(\hat{\gamma}^{(s)}; \mathcal{D}_n), \quad \hat{\gamma} = \hat{\gamma}^{(s^*)}.$$

Why multi-start?

The LOOCV objective may be non-convex, so multiple initial values reduce sensitivity to a single local optimum.



Contents

- 1 Motivation
- 2 Methodology
- 3 Package Usage**
- 4 Examples and Applications
- 5 Extensions
- 6 Summary



fit_BKP() — Data Inputs

```
fit_BKP(X, y, m, Xbounds = NULL,  
prior = "noninformative", r0 = 2, p0 = mean(y / m),  
kernel = "gaussian", isotropic = TRUE, theta = NULL,  
loss = "brier", ess = "none", n_multi_start = NULL, n_threads = 1)
```

- The first group specifies the core **data inputs**:
 - X: an $n \times d$ numeric input matrix or data frame
 - y: a length- n vector of observed success counts
 - m: a length- n vector of corresponding binomial trial counts
 - Xbounds: an optional $d \times 2$ matrix of lower and upper bounds, used to normalize X to $[0, 1]^d$ before fitting. If NULL, X is assumed to be already normalized.



fit_BKP() — Prior Specification

```
fit_BKP(X, y, m, Xbounds = NULL,  
        prior = "noninformative", r0 = 2, p0 = mean(y / m),  
        kernel = "gaussian", isotropic = TRUE, theta = NULL,  
        loss = "brier", ess = "none", n_multi_start = NULL, n_threads = 1)
```

- The second group specifies the **prior distribution**:
 - prior: selects "noninformative" (default), "fixed", or "adaptive"
 - the prior is constructed using the get_prior() function
 - r0 and p0: global prior precision and mean



fit_BKP() — Kernel Specification

```
fit_BKP(X, y, m, Xbounds = NULL,  
prior = "noninformative", r0 = 2, p0 = mean(y / m),  
kernel = "gaussian", isotropic = TRUE, theta = NULL,  
loss = "brier", ess = "none", n_multi_start = NULL, n_threads = 1)
```

- The third group specifies the **kernel and length-scale structure**:
 - kernel: "gaussian" (default), "matern52", "matern32", or "wendland"
 - isotropic: TRUE uses one shared length scale; FALSE uses dimension-specific length scales
 - theta: a positive scalar for isotropic kernels, or a scalar/length- d vector for anisotropic kernels
 - If `theta = NULL`, it is selected by LOOCV; otherwise, the supplied value is used and optimization is skipped



fit_BKP() — Tuning and Computation

```
fit_BKP(X, y, m, Xbounds = NULL,  
prior = "noninformative", r0 = 2, p0 = mean(y / m),  
kernel = "gaussian", isotropic = TRUE, theta = NULL,  
loss = "brier", ess = "none", n_multi_start = NULL, n_threads = 1)
```

- The final group controls **hyperparameter tuning, ESS calibration, and computation**:
 - loss: "brier" (default) or "log_loss" for LOOCV tuning
 - ess: "none" uses the standard BKP update; "shepard" calibrates the kernel-weighted data precision
 - n_multi_start: number of optimization starts; if NULL, the default is 10 for isotropic kernels and $10d$ for anisotropic kernels
 - n_threads: number of threads used in multi-start optimization; relevant only when $\theta = \text{NULL}$



`fit_BKP()` — Returned Object

```
BKPmodel <- fit_BKP(X, y, m, Xbounds = Xbounds)
```

- The returned BKP object stores:
 - the fitted kernel, length-scale parameters, and LOOCV results
 - the prior specification and normalized training data
 - the prior parameters α_0 and β_0
 - the posterior parameters α_n and β_n
- The fitted object supports standard S3 methods, including `summary()`, `predict()`, `plot()`, `simulate()`, and `print()`.



Contents

- 1 Motivation
- 2 Methodology
- 3 Package Usage
- 4 Examples and Applications**
- 5 Extensions
- 6 Summary



Contents

4 Examples and Applications

- Synthetic Examples
- Real-Data Application



Logistic Probability Curve

- Let $x \in [-2, 2]$, and define the true success-probability function as

$$\pi_1(x) = \frac{1}{1 + e^{-3x}}.$$

- We generate binomial observations at $n = 7$ input locations.
- This monotone logistic curve provides a simple benchmark for demonstrating the standard BKP workflow.
- [Reproduction code](#)



Model Fitting

```
# Define true success probability function
true_pi_fun <- function(x) {
  1/(1+exp(-3*x))
}

# Dataset
n <- 7
Xbounds <- matrix(c(-2,2), nrow = 1)
X <- lhs(n = n, rect = Xbounds)
true_pi <- true_pi_fun(X)
m <- sample(100, n, replace = TRUE)
y <- rbinom(n, size = m, prob = true_pi)

# Fit BKP model
BKP_model_1D_1 <- fit_BKP(X, y, m, Xbounds = Xbounds)
```



Fitted Model Summary

```
summary(BKP_model_1D_1)
```

```
##
##          Beta Kernel Process (BKP) Model
##
## Number of observations (n):  7
## Input dimensionality (d):    1
## Kernel type:                  (isotropic) gaussian
## Optimized kernel parameters: 0.1748
## Minimum achieved loss:       0.01165
## Loss function:                brier
## Prior type:                   noninformative
##
## Posterior predictive summary (training points):
##               Mean Median      SD      Min      Max
## Posterior means      0.5328 0.7098 0.4079 0.0625 0.9407
## Posterior variances 0.0013 0.0011 0.0007 0.0007 0.0027
```



Prediction at New Inputs

```
Xnew <- lhs(n = 10, rect = Xbounds)
predict(BKP_model_1D_1, Xnew)
```

```
## Prediction results on new data (Xnew).
```

```
## Total number of prediction points: 10
```

```
##
```

```
## Preview of predictions for new data (first 6 of 10 points):
```

```
##           x      mean variance 2.5% quantile 97.5% quantile
```

```
## -0.1791 0.4512    0.0033      0.3403      0.5646
```

```
## -1.5726 0.0570    0.0007      0.0176      0.1173
```

```
##  1.4394 0.9449    0.0009      0.8714      0.9883
```

```
##  0.6069 0.8517    0.0015      0.7673      0.9198
```

```
##  1.8955 0.9365    0.0017      0.8355      0.9908
```

```
##  0.8400 0.8983    0.0011      0.8230      0.9545
```

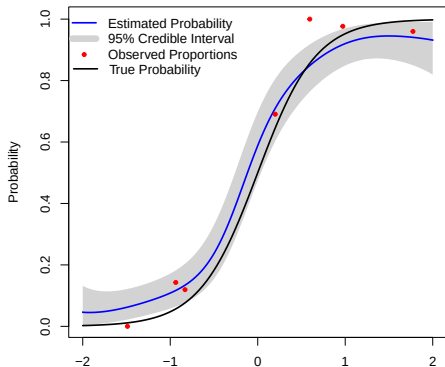
```
## ...
```



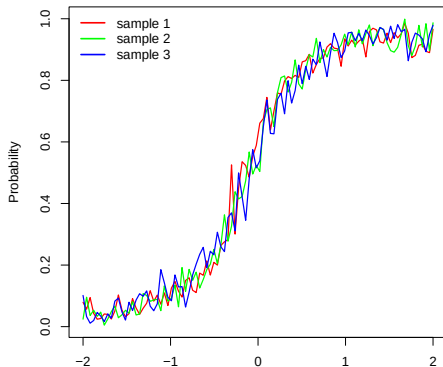
Posterior Inference and Simulation

```
plot(BKP_model_1D_1)
Xnew <- matrix(seq(-2, 2, length = 100), ncol = 1)
sim <- simulate(BKP_model_1D_1, nsim = 3, Xnew = Xnew)
```

Estimated Probability



Simulated Probability Curves





Highly Nonlinear Probability Curve

- Unlike the monotone logistic curve in Example 1, this probability function is non-monotone and locally oscillatory.
- For $x \in [-2, 2]$, define

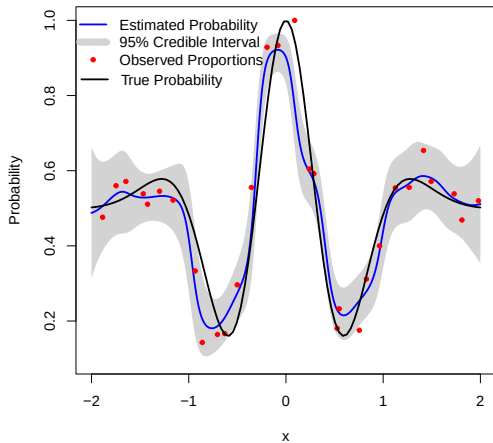
$$\pi_2(x) = \frac{1}{2} \left[1 + e^{-x^2} \cos \left(10 \frac{1 - e^{-x}}{1 + e^{-x}} \right) \right],$$

- We generate $n = 30$ binomial observations and fit the default BKP model.
- [Reproduction code](#)

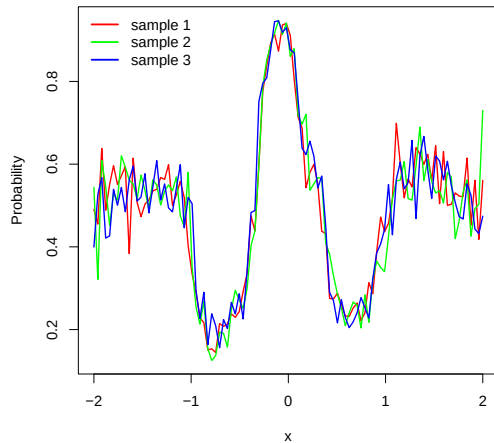


Posterior Inference for the Nonlinear Curve

Estimated Probability



Simulated Probability Curves

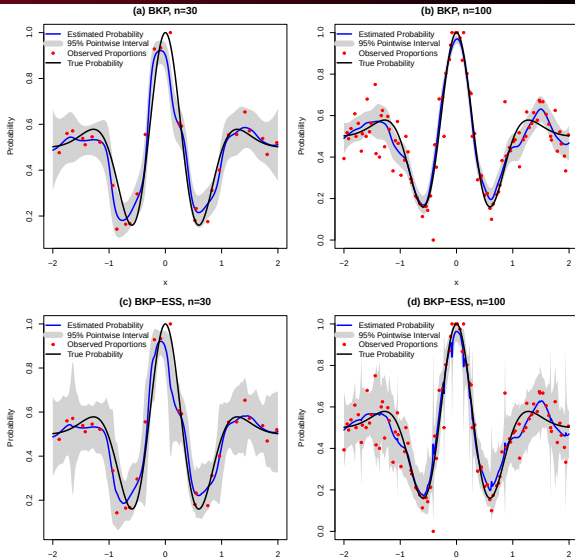


[Explore the interactive example](#)



Effect of ESS Calibration on Pointwise Intervals

Evaluation set	Sample size	Model	Coverage (%)
Evaluation grid	$n = 30$	BKP	87.1 (6.4)
		BKP-ESS	94.1 (3.1)
	$n = 100$	BKP	90.9 (4.1)
		BKP-ESS	97.6 (1.0)
Training locations	$n = 30$	BKP	88.1 (6.7)
		BKP-ESS	94.6 (3.3)
	$n = 100$	BKP	90.8 (4.1)
		BKP-ESS	97.6 (1.0)





Two-Dimensional Probability Surface

- Let $\boldsymbol{x} \in [0, 1]^2$.
- A nonlinear latent surface $f(\boldsymbol{x})$ is constructed from a rescaled Goldstein–Price function (Picheny et al., 2013).
- The true success probability is

$$\pi(\boldsymbol{x}) = \Phi\{f(\boldsymbol{x})\},$$

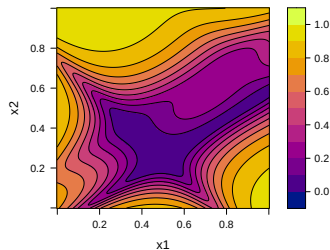
where $\Phi(\cdot)$ is the standard normal cumulative distribution function.

- We generate $n = 100$ binomial observations and fit the BKP model.
- [Reproduction code](#)

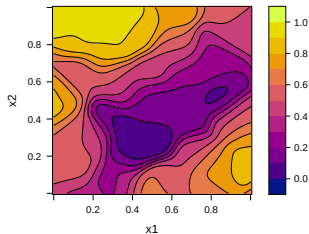


Posterior Inference for the Two-Dimensional Surface

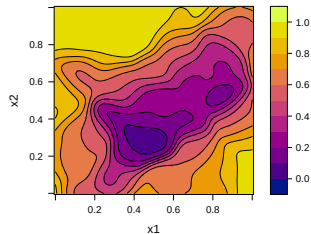
True Probability



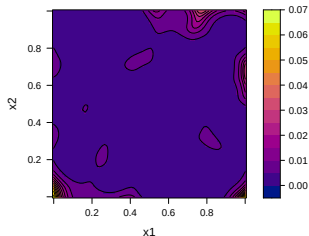
Posterior Mean



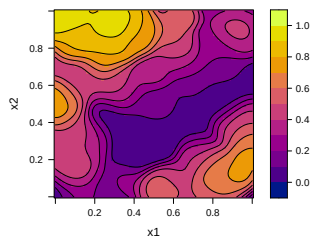
95% CI Upper



Posterior Variance



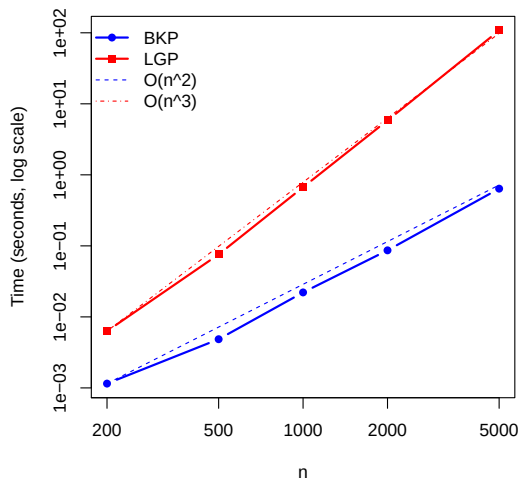
95% CI Lower



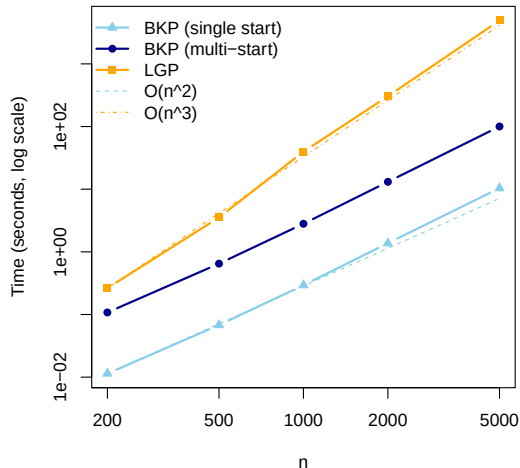


Comparison of Computation Times

Fixed Hyperparameter



Optimization-based Methods





Contents

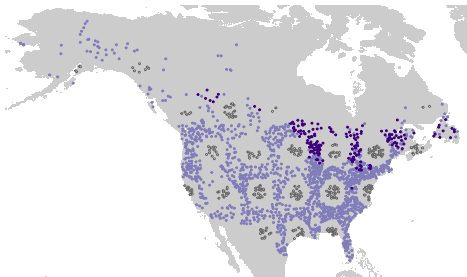
4 Examples and Applications

- Synthetic Examples
- Real-Data Application



Mourning Warbler Distribution Modeling

Distribution data – Mourning Warbler

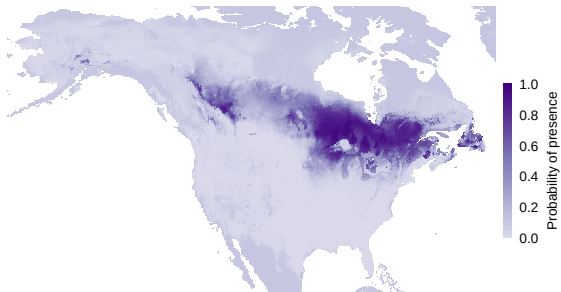


- Presence–absence data from the North American Breeding Bird Survey (Sauer et al., 2014)
- 1,613 training observations and 301 spatially withheld test observations
- Eight continuous bioclimatic predictors
- Longitude and latitude are used only for spatial partitioning and visualization
- The response is Bernoulli, so $m_i = 1$ for every observation

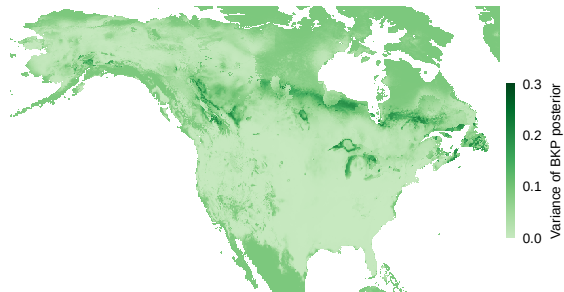
Reproduction code

Predicted Presence Probability and Posterior Uncertainty

(a) BKP – Predicted distribution



(b) BKP – Prediction uncertainty





Contents

- 1 Motivation
- 2 Methodology
- 3 Package Usage
- 4 Examples and Applications
- 5 Extensions**
- 6 Summary



Contents

5 Extensions

- TwinBKP
- DKP



Scalable BKP via a Global–Local Approximation

Full BKP

Full BKP uses all n observations at every input location.

- Tuning cost: $\mathcal{O}(Tn^2)$
- Memory: $\mathcal{O}(n^2)$

Global–local idea

- The global subset provides broad coverage of the input space.
- Prediction-specific local neighbours refine the posterior near the target location.
- ref: Vakayil and Joseph (2022, 2024)

TwinBKP subsets

A representative global subset is shared across all prediction locations:

$$G \subset \{1, \dots, n\}, \quad |G| = g.$$

For each target $\mathbf{x}$, select ℓ nearest non-global observations:

$$L(\mathbf{x}) \subset \{1, \dots, n\} \setminus G, \quad |L(\mathbf{x})| = \ell.$$



Two-Stage Conjugate Update

TwinBKP posterior

$$\pi(\mathbf{x}) \mid \mathcal{D}_{U(\mathbf{x})} \sim \text{Beta}(\alpha_U(\mathbf{x}), \beta_U(\mathbf{x})), \quad U(\mathbf{x}) = G \cup L(\mathbf{x}).$$

Stage 1: Global update

The global subset defines an intermediate Beta distribution:

$$\begin{aligned} \alpha_G(\mathbf{x}) &= \alpha_0(\mathbf{x}) + \sum_{i \in G} k_g(\mathbf{x}, \mathbf{x}_i; \theta_g) y_i, \\ \beta_G(\mathbf{x}) &= \beta_0(\mathbf{x}) + \sum_{i \in G} k_g(\mathbf{x}, \mathbf{x}_i; \theta_g) (m_i - y_i). \end{aligned}$$

Stage 2: Local refinement

The intermediate distribution is updated using the local neighbours:

$$\begin{aligned} \alpha_U(\mathbf{x}) &= \alpha_G(\mathbf{x}) + \sum_{i \in L(\mathbf{x})} k_\ell(\mathbf{x}, \mathbf{x}_i; \theta_\ell) y_i, \\ \beta_U(\mathbf{x}) &= \beta_G(\mathbf{x}) + \sum_{i \in L(\mathbf{x})} k_\ell(\mathbf{x}, \mathbf{x}_i; \theta_\ell) (m_i - y_i). \end{aligned}$$



Kernel Tuning and Computational Scaling

Default subset sizes

$$g = \min \{n - 1, 50d, \max(\lfloor \sqrt{n} \rfloor, 10d)\}, \quad \ell = \min \{n - g, \max(25, 3d)\}.$$

Kernel parameters

- The global length scale θ_g is selected by the same LOOCV criterion as full BKP, but using only the global subset.
- The scalar local range parameter is not separately optimized. It is set to the empirical covering radius

$$\hat{\theta}_\ell = \max_{1 \leq i \leq n} \min_{j \in G} \|\mathbf{x}_i - \mathbf{x}_j\|_2$$

on the normalized input scale.

Computational cost

Fitting: $\mathcal{O}(n \log n + T_g g^2)$,

Memory: $\mathcal{O}(n + g^2)$,

Prediction at t locations: $\mathcal{O}\{t(\log n + g + \ell)\}$.

No dense $n \times n$ kernel matrix is stored by default.



Large- n Illustration: TwinBKP

```
TwinBKP_model <- fit_TwinBKP(X, y, m, Xbounds = Xbounds)
```

The nonlinear probability function from Example 2 is reused, with the sample size increased from 30 to 500.

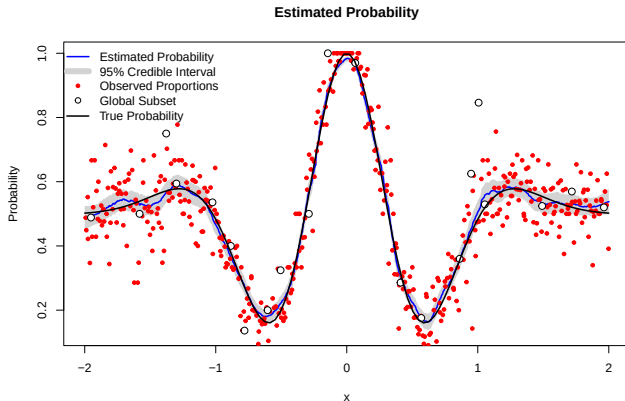
Realized default structure

$$g = 22$$

$$\ell = 25$$

Global kernel: Gaussian

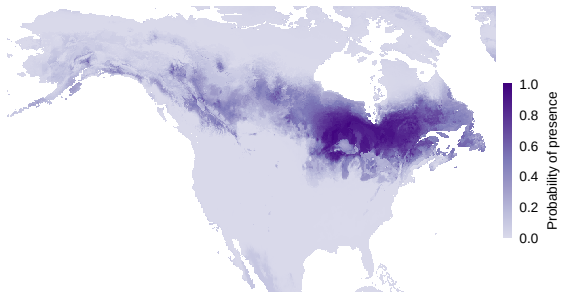
Local kernel: Wendland



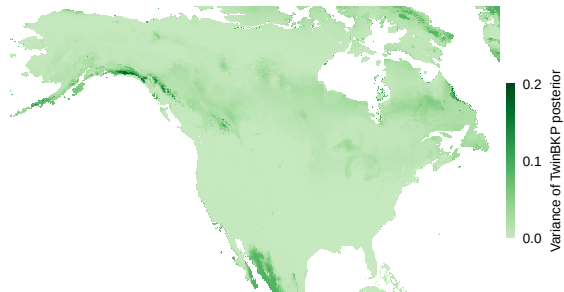
[Explore the interactive TwinBKP example](#)

TwinBKP Prediction for Mourning Warbler

(c) TwinBKP – Predicted distribution

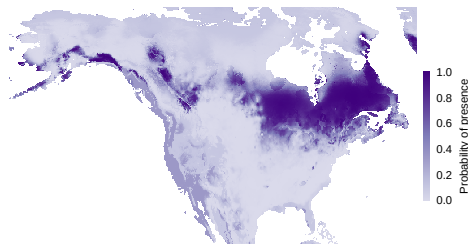


(d) TwinBKP – Prediction uncertainty

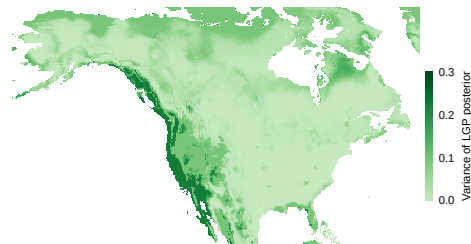


Mourning Warbler: BKP versus LGP

(e) LGP – Predicted distribution



(f) LGP – Prediction uncertainty



Model	AUC	Brier score	Elapsed time (s)
BKP	0.888	0.0691	9.04
TwinBKP	0.913	0.0674	0.05
LGP (Golding and Purse, 2016)	0.852	0.0798	541.82



Contents

5 Extensions

- TwinBKP
- DKP



Extension to Multiclass Responses: DKP

Key idea

Binomial response $\longrightarrow$ Multinomial response

Beta distribution $\longrightarrow$ Dirichlet distribution

$$\pi(\mathbf{x}) \longrightarrow \boldsymbol{\pi}(\mathbf{x}) = \{\pi_1(\mathbf{x}), \dots, \pi_q(\mathbf{x})\}$$

Kernel-weighted Dirichlet update

Let $\mathbf{Y} \in \mathbb{R}^{n \times q}$ contain the observed class counts. Then

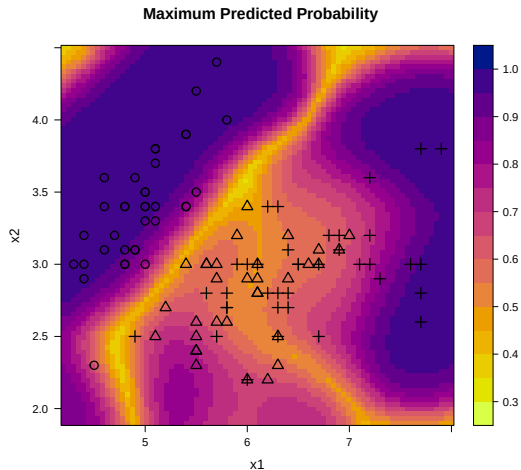
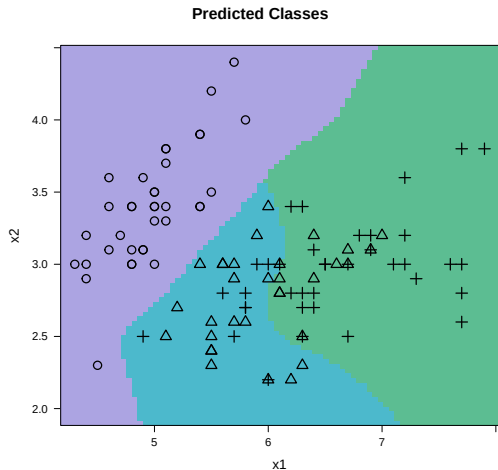
$$\boldsymbol{\pi}(\mathbf{x}) \mid \mathcal{D}_n \sim \text{Dirichlet}(\boldsymbol{\alpha}_n(\mathbf{x})),$$

$$\boldsymbol{\alpha}_n(\mathbf{x}) = \boldsymbol{\alpha}_0(\mathbf{x}) + \mathbf{k}(\mathbf{x})^\top \mathbf{Y}.$$

DKP extends the BKP update from a scalar success probability to an input-dependent vector of class probabilities.



Multiclass Illustration: Iris Classification



[Reproduction code](#)



Contents

- 1 Motivation
- 2 Methodology
- 3 Package Usage
- 4 Examples and Applications
- 5 Extensions
- 6 Summary**

Summary

One Grammar, Four Models

Response type	Full-data model	Global–local approximation
Binary or binomial	<code>fit_BKP()</code>	<code>fit_TwinBKP()</code>
Categorical or multinomial	<code>fit_DKP()</code>	<code>fit_TwinDKP()</code>

Computational Profile

Method	Inference	Fitting / tuning	Memory	Pointwise prediction
BKP	Direct,	$\mathcal{O}(Tn^2)$	$\mathcal{O}(n^2)$	$\mathcal{O}(n)$
TwinBKP	closed-form	$\mathcal{O}(n \log n + T_g g^2)$	$\mathcal{O}(n + g^2)$	$\mathcal{O}(\log n + g + \ell)$
Dense LGP	Latent, approximate	$\mathcal{O}(T_L n^3)$	$\mathcal{O}(n^2)$	$\mathcal{O}(n^2)$



Future Research Directions

Theory

- Consistency and convergence rates for full BKP and DKP
- Approximation-error bounds for the Twin variants

Calibration

- Finite-sample 95% marginal coverage for future responses
- Conditional and simultaneous calibration across the input space

Broader modeling scope

- Longitudinal, spatio-temporal, functional, and multivariate data
- Over-dispersed counts and other non-binomial outcome families



Resources

Software

-  BKP on CRAN: package and documentation
-  Source code and package development

Paper and project materials

-  BKP software paper on arXiv
-  Code, data, results, figures, slides, and manuscript

Project website

Interactive examples, installation guidance, and resources



Scan to visit the BKP website
jiangyan-zhao.github.io/BKP-website



- Fan, J., M. Farmen, and I. Gijbels (1998). Local maximum likelihood estimation and inference. *Journal of the Royal Statistical Society Series B: Statistical Methodology* 60(3), 591–608.
- Goetschalckx, R., P. Poupart, and J. Hoey (2011). Continuous correlated beta processes. In *Proceedings of the Twenty-Second International Joint Conference on Artificial Intelligence - Volume Two*, IJCAI'11, pp. 1269–1274. AAAI Press.
- Golding, N. and B. V. Purse (2016). Fast and flexible Bayesian species distribution modelling using Gaussian processes. *Methods in Ecology and Evolution* 7, 598–608.
- MacKenzie, C. A., T. B. Trafalis, and K. Barker (2014). A Bayesian beta kernel model for binary classification and online learning problems. *Statistical Analysis and Data Mining: The ASA Data Science Journal* 7(6), 434–449.
- Nickisch, H. and C. E. Rasmussen (2008). Approximations for binary Gaussian process classification. *Journal of Machine Learning Research* 9(67), 2035–2078.
- Picheny, V., T. Wagner, and D. Ginsbourger (2013). A benchmark of Kriging-based infill criteria for noisy optimization. *Structural and Multidisciplinary Optimization* 48(3), 607–626.



- Rasmussen, C. E. and C. K. Williams (2006). *Gaussian Processes for Machine Learning*. Cambridge, MA: the MIT Press.
- Rolland, P., A. Kavis, A. Immer, A. Singla, and V. Cevher (2019). Efficient learning of smooth probability functions from Bernoulli tests with guarantees. In K. Chaudhuri and R. Salakhutdinov (Eds.), *Proceedings of the 36th International Conference on Machine Learning*, Volume 97 of *Proceedings of Machine Learning Research*, pp. 5459–5467. PMLR.
- Sauer, J., J. Hines, J. Fallon, K. Pardieck, D. Ziolkowski Jr, and W. Link (2014). The north american breeding bird survey, results and analysis 1966–2012. version 02.19.2014. Technical report, USGS Patuxent Wildlife Research Center, Laurel, MD.
- Shepard, D. (1968). A two-dimensional interpolation function for irregularly-spaced data. In *Proceedings of the 1968 23rd ACM National Conference*, pp. 517–524.
- Vakayil, A. and V. R. Joseph (2022). Data twinning. *Statistical Analysis and Data Mining: The ASA Data Science Journal* 15(5), 598–610.
- Vakayil, A. and V. R. Joseph (2024). A global-local approximation framework for large-scale Gaussian process modeling. *Technometrics* 66(2), 295–305.